

Refraction at a spherical surface

① for Convex Spherical surface -

In $\triangle ONC$

$$i = \alpha + \beta \text{ --- (1) (external Angle property)}$$

In $\triangle NCI$

$$\beta = r + \gamma \text{ (external angle property)}$$

$$\Rightarrow r = \beta - \gamma \text{ --- (2)}$$

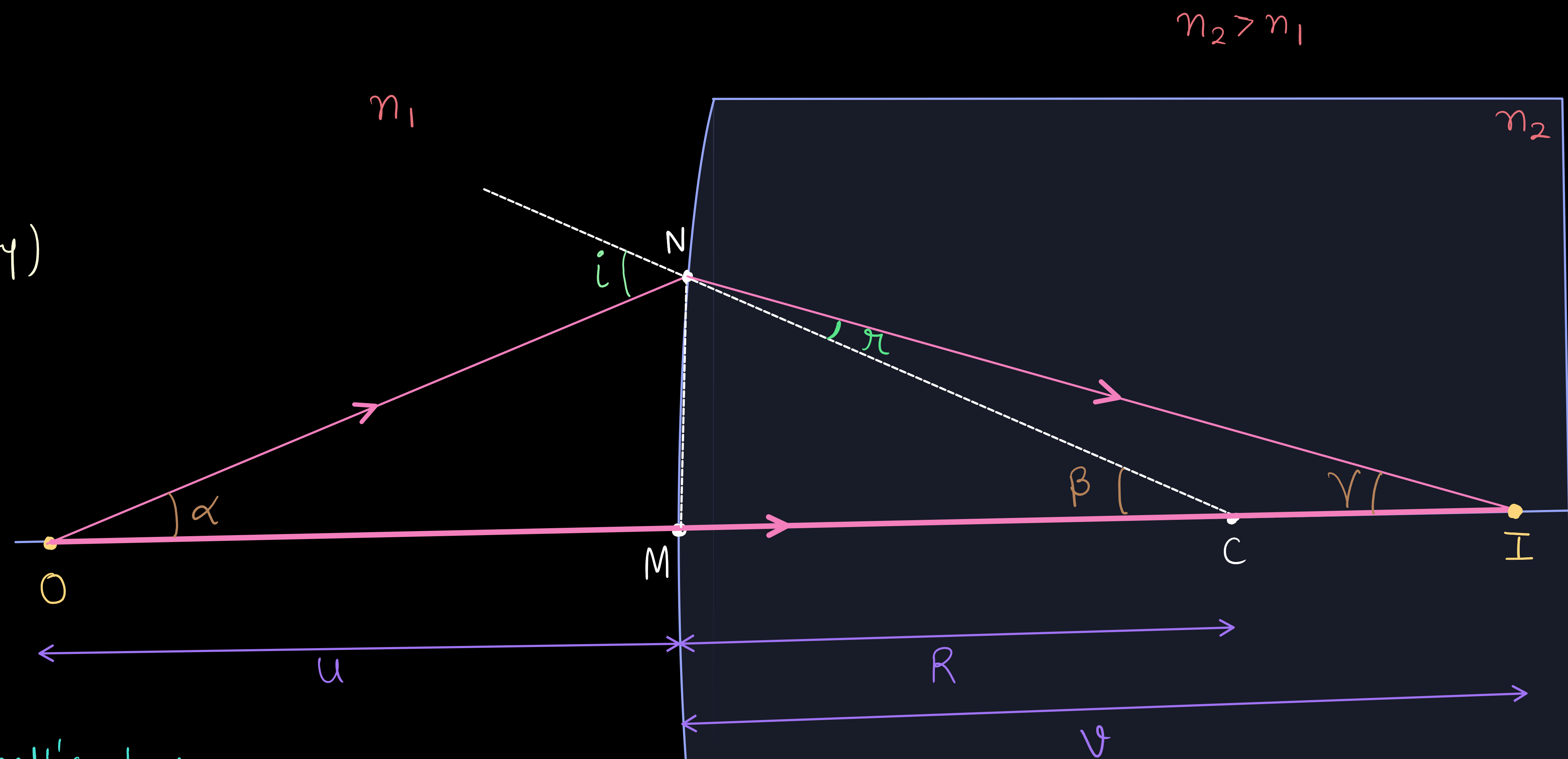
for paraxial rays -

i, r, α, β and γ are small

$$\therefore \tan \alpha \approx \alpha = \frac{NM}{OM} \text{ --- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MC} \text{ --- (4)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MI} \text{ --- (5)}$$



Now, using Snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$n_1 i = n_2 r \text{ --- (6) } (\because \sin i \approx i, \sin r \approx r)$$

$$n_1 (\alpha + \beta) = n_2 (\beta - \gamma)$$

Substitute values of α, β and γ from eqⁿ (3), (4) & (5)

$$n_1 \left(\frac{NM}{OM} + \frac{NM}{MC} \right) = n_2 \left(\frac{NM}{MC} - \frac{NM}{MI} \right)$$

Refraction at a spherical surface

$$n_1 \left(\frac{\cancel{NM}}{OM} + \frac{\cancel{NM}}{MC} \right) = n_2 \left(\frac{\cancel{NM}}{MC} - \frac{\cancel{NM}}{MI} \right)$$

$$n_1 \left(\frac{1}{OM} + \frac{1}{MC} \right) = n_2 \left(\frac{1}{MC} - \frac{1}{MI} \right)$$

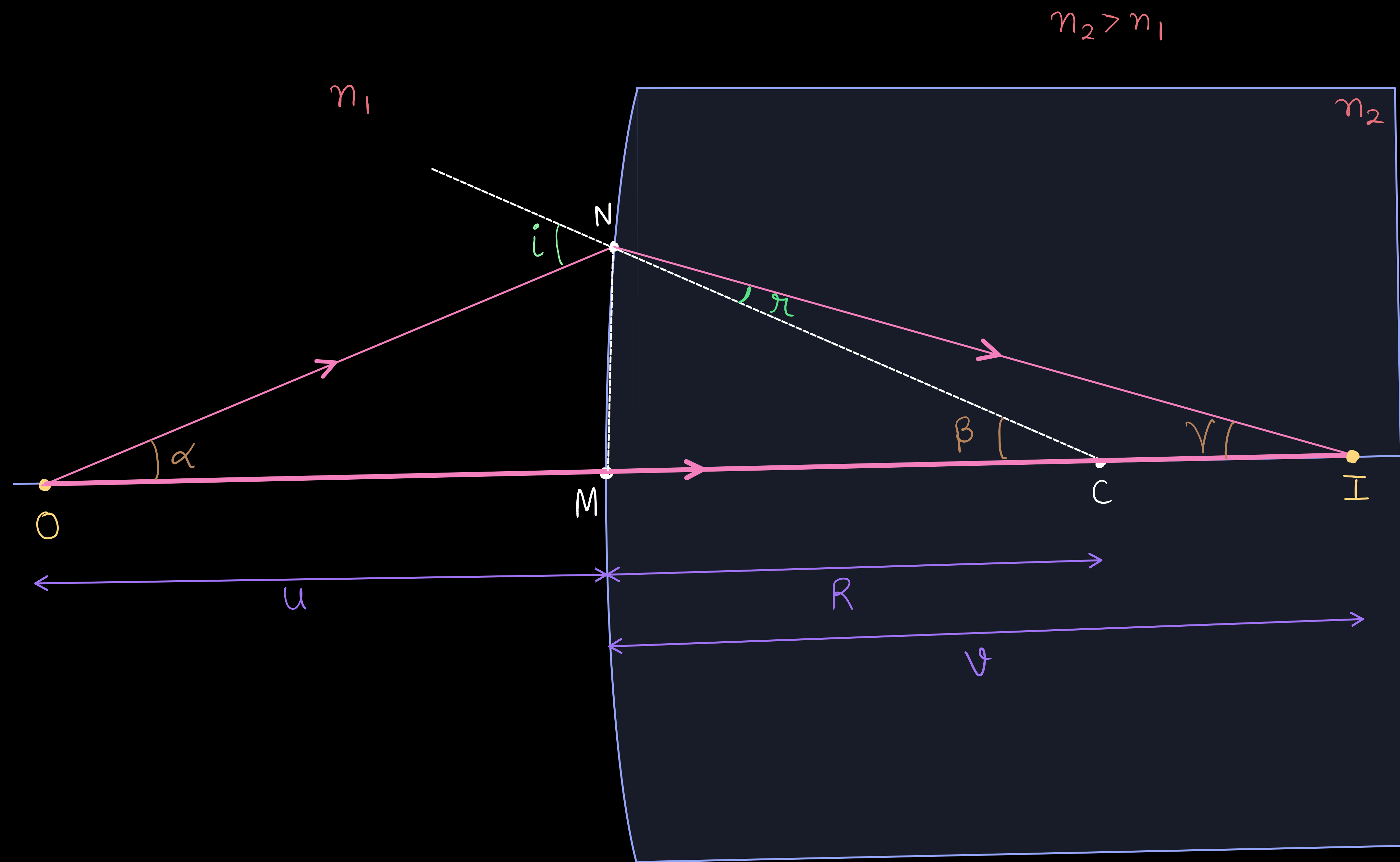
$$\frac{n_1}{OM} + \frac{n_1}{MC} = \frac{n_2}{MC} - \frac{n_2}{MI}$$

$$\frac{n_1}{OM} + \frac{n_2}{MI} = \frac{(n_2 - n_1)}{MC}$$

Now, using sign convention -

$$OM = -u, \quad MI = v, \quad MC = R$$

$$\boxed{-\frac{n_1}{u} + \frac{n_2}{v} = \frac{(n_2 - n_1)}{R}} \quad \text{***}$$



Refraction at a spherical surface

② for convex surface (Virtual image)

In $\triangle NOC$

$$i = \beta + \gamma \quad \text{--- (1) (external angle)}$$

In $\triangle NIC$

$$r = \alpha + \gamma \quad \text{--- (2) (external angle)}$$

for paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MI} \quad \text{--- (3)}$$

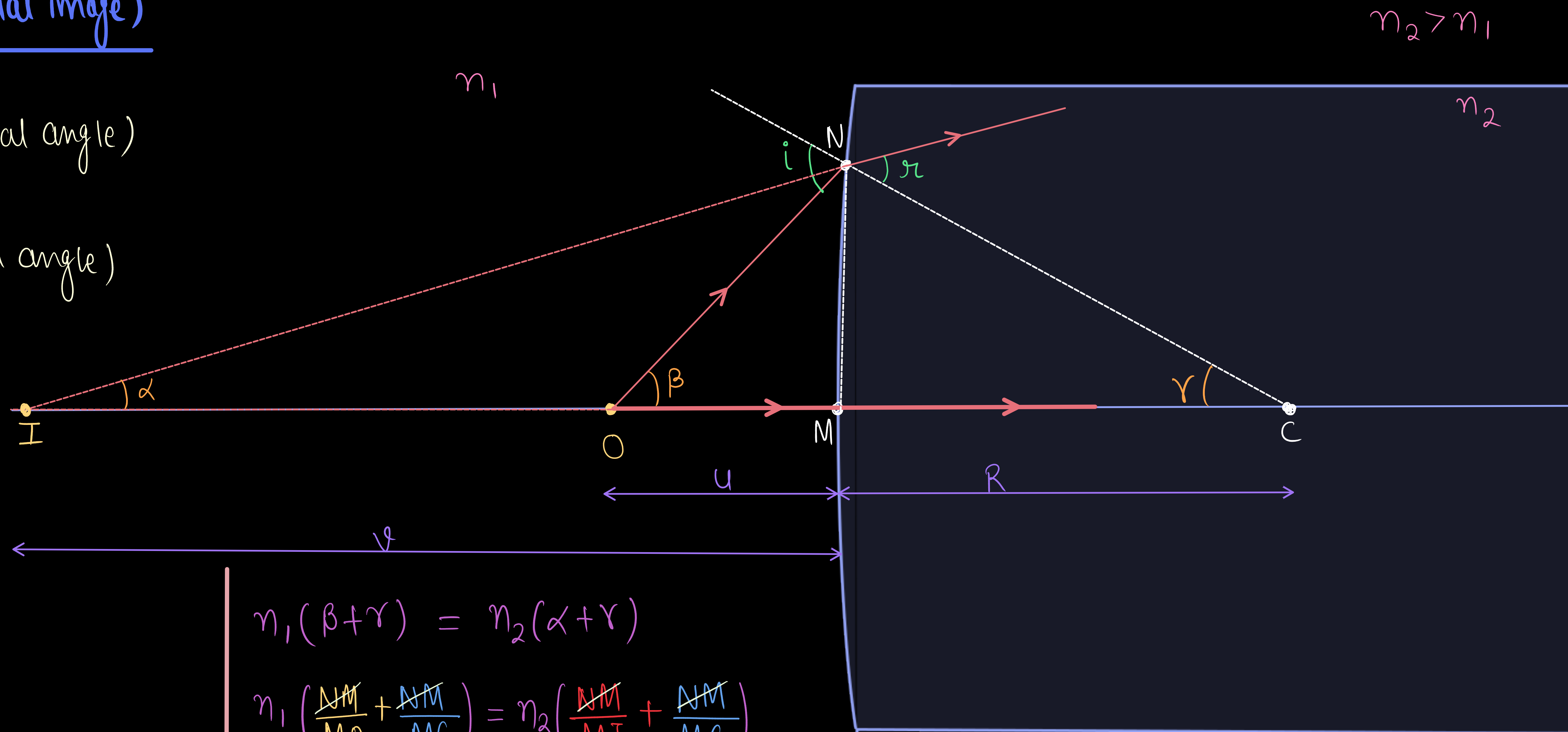
$$\tan \beta \approx \beta = \frac{NM}{MO} \quad \text{--- (4)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MC} \quad \text{--- (5)}$$

Now, using Snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$\text{or } n_1 i = n_2 r \quad (\because \sin i \approx i, \sin r \approx r)$$



$$n_1(\beta + \gamma) = n_2(\alpha + \gamma)$$

$$n_1 \left(\frac{NM}{MO} + \frac{NM}{MC} \right) = n_2 \left(\frac{NM}{MI} + \frac{NM}{MC} \right)$$

$$n_1 \left(\frac{1}{MO} + \frac{1}{MC} \right) = n_2 \left(\frac{1}{MI} + \frac{1}{MC} \right)$$

$$\frac{n_1}{MO} + \frac{n_1}{MC} = \frac{n_2}{MI} + \frac{n_2}{MC}$$

$$\frac{n_1}{MO} - \frac{n_2}{MI} = \frac{n_2 - n_1}{MC} \quad \text{--- (6)}$$

using sign convention $\rightarrow MO = -u, MI = -v, MC = R$

$$\boxed{-\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}}$$

Refraction at a spherical surface

③ for convex surface (Object is in denser Medium)

In $\triangle NCO$ -

$$\beta = i + \gamma \text{ (external angle)}$$

$$\text{or } i = \beta - \gamma \text{ --- (1)}$$

In $\triangle NIC$ -

$$r = \alpha + \beta \text{ (ext. angle)}$$

for paraxial rays -

$$\sin \alpha \approx \alpha = \frac{NM}{MI} \text{ --- (3)}$$

$$\sin \beta \approx \beta = \frac{NM}{MC} \text{ --- (4)}$$

$$\sin \gamma \approx \gamma = \frac{NM}{MO} \text{ --- (5)}$$

using snell's law -

$$n_2 \sin i = n_1 \sin r$$

$$n_2 i = n_1 r$$

$$n_2 (\beta - \gamma) = n_1 (\alpha + \beta)$$

$$n_2 \left(\frac{NM}{MC} - \frac{NM}{MO} \right) = n_1 \left(\frac{NM}{MI} + \frac{NM}{MC} \right)$$

$$n_2 \left(\frac{1}{MC} - \frac{1}{MO} \right) = n_1 \left(\frac{1}{MI} + \frac{1}{MC} \right)$$

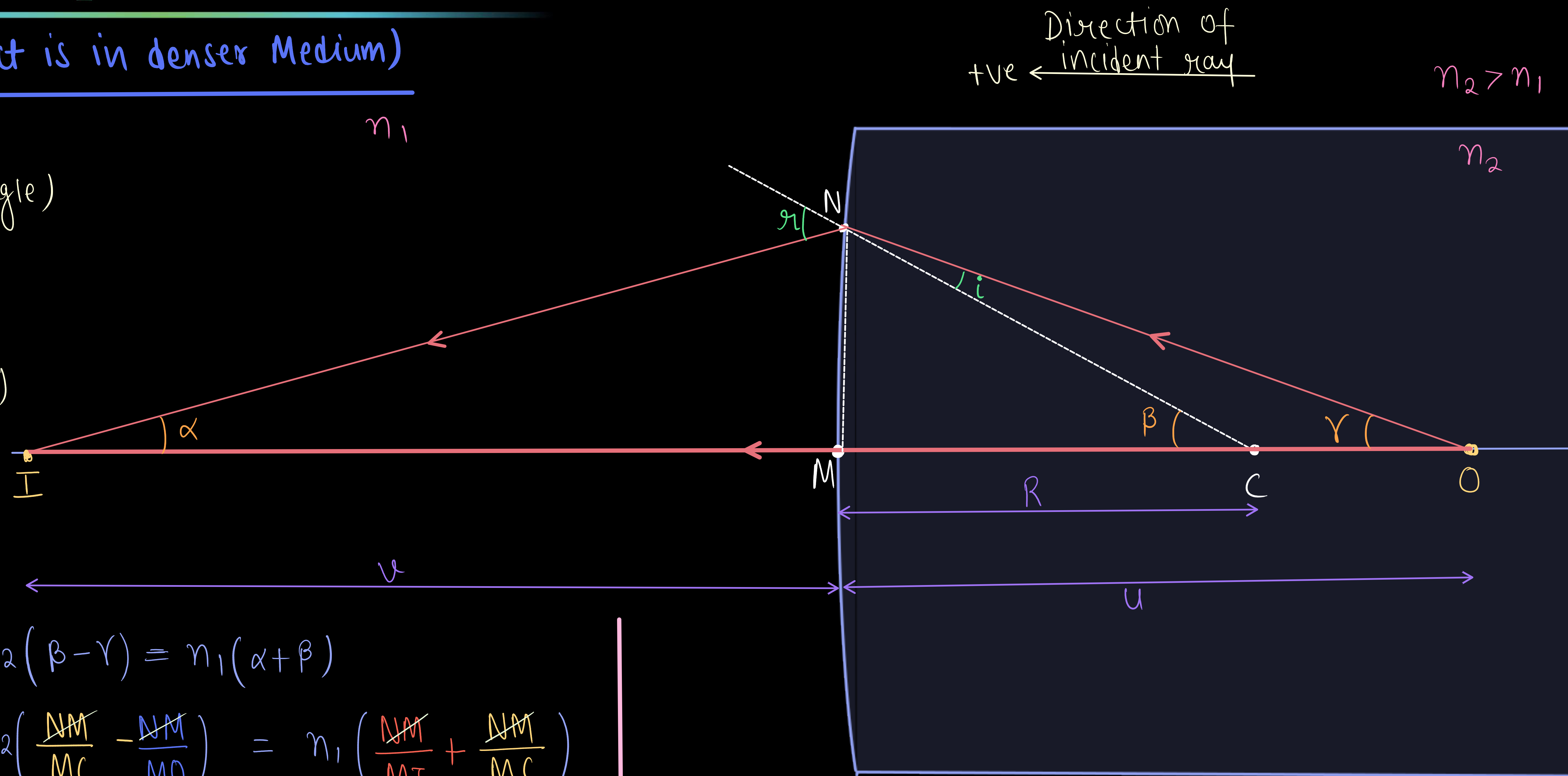
$$\frac{n_2}{MC} - \frac{n_2}{MO} = \frac{n_1}{MI} + \frac{n_1}{MC}$$

$$\frac{n_2 - n_1}{MC} = \frac{n_1}{MI} + \frac{n_2}{MO}$$

using sign convention - $MC = -R$, $MO = -u$, $MI = v$

$$\frac{n_2 - n_1}{-R} = \frac{n_1}{v} - \frac{n_2}{u} \text{ or}$$

$$\boxed{\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}}$$



Refraction at a spherical surface

④ for Concave spherical surface -

$$n_2 > n_1$$

In $\triangle NCO$

$$\gamma = \alpha + i \text{ (ext. Ang.)}$$

$$\text{or } i = \gamma - \alpha \text{ --- (1)}$$

In $\triangle NCI$

$$\gamma = \beta + r \text{ (ext. Ang.)}$$

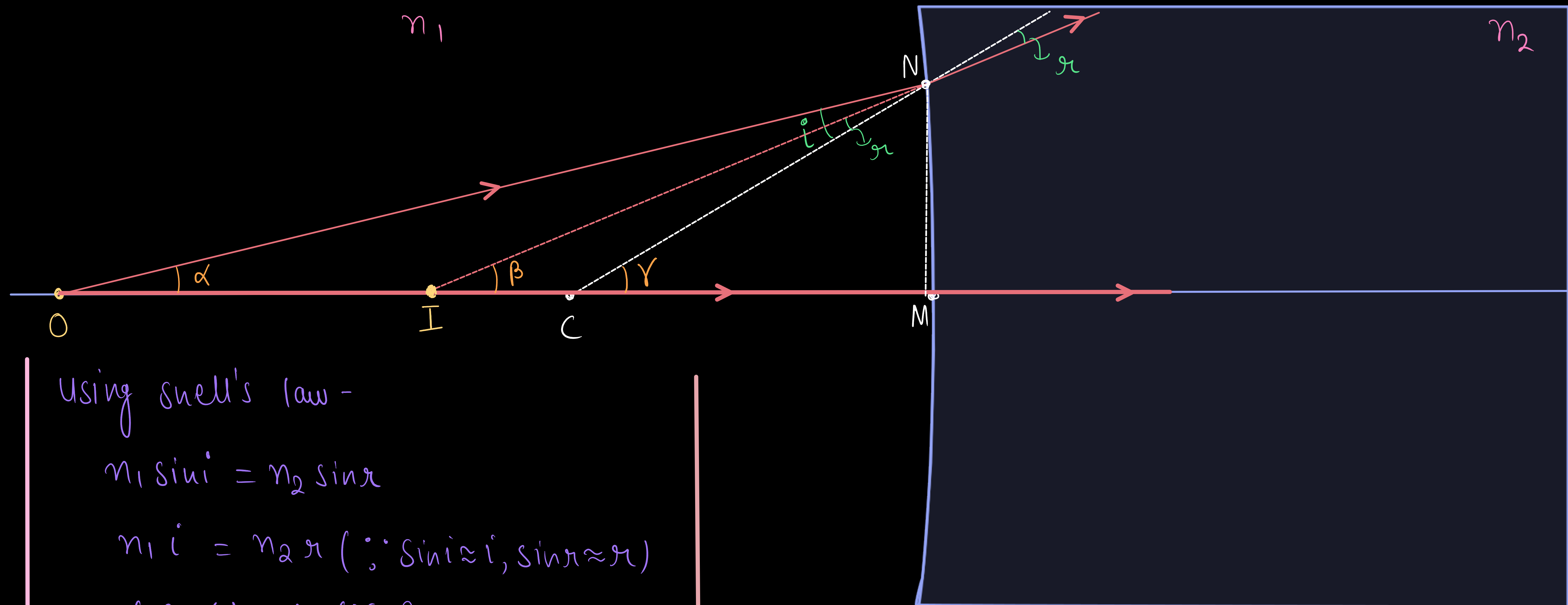
$$\text{or } r = \gamma - \beta \text{ --- (2)}$$

for paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MO} \text{ --- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MI} \text{ --- (3)}$$

$$\tan \gamma \approx \gamma = \frac{NM}{MC} \text{ --- (4)}$$



using snell's law -

$$n_1 \sin i = n_2 \sin r$$

$$n_1 i = n_2 r \text{ (} \because \sin i \approx i, \sin r \approx r \text{)}$$

$$n_1 (\gamma - \alpha) = n_2 (\gamma - \beta)$$

$$n_1 \left(\frac{NM}{MC} - \frac{NM}{MO} \right) = n_2 \left(\frac{NM}{MC} - \frac{NM}{MI} \right)$$

$$\frac{n_1}{MC} - \frac{n_1}{MO} = \frac{n_2}{MC} - \frac{n_2}{MI}$$

$$\frac{n_2}{MI} - \frac{n_1}{MO} = \frac{n_2 - n_1}{MC}$$

using sign convention, $MI = -v$, $MO = -u$, $MC = -R$

$$\frac{n_2}{-v} - \frac{n_1}{-u} = \frac{n_2 - n_1}{-R} \text{ or}$$

$$\boxed{\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}}$$

Refraction at a spherical surface

⑤ for concave surface (object in denser Medium)

In ΔNCO -

$$i = \alpha + \gamma \quad \text{--- (1) (ext. Angle)}$$

In ΔNCI -

$$r = \alpha + \beta \quad \text{--- (2) (ext. Angle)}$$

For paraxial rays -

$$\tan \alpha \approx \alpha = \frac{NM}{MC} \quad \text{--- (3)}$$

$$\tan \beta \approx \beta = \frac{NM}{MI} \quad \text{--- (4)}$$

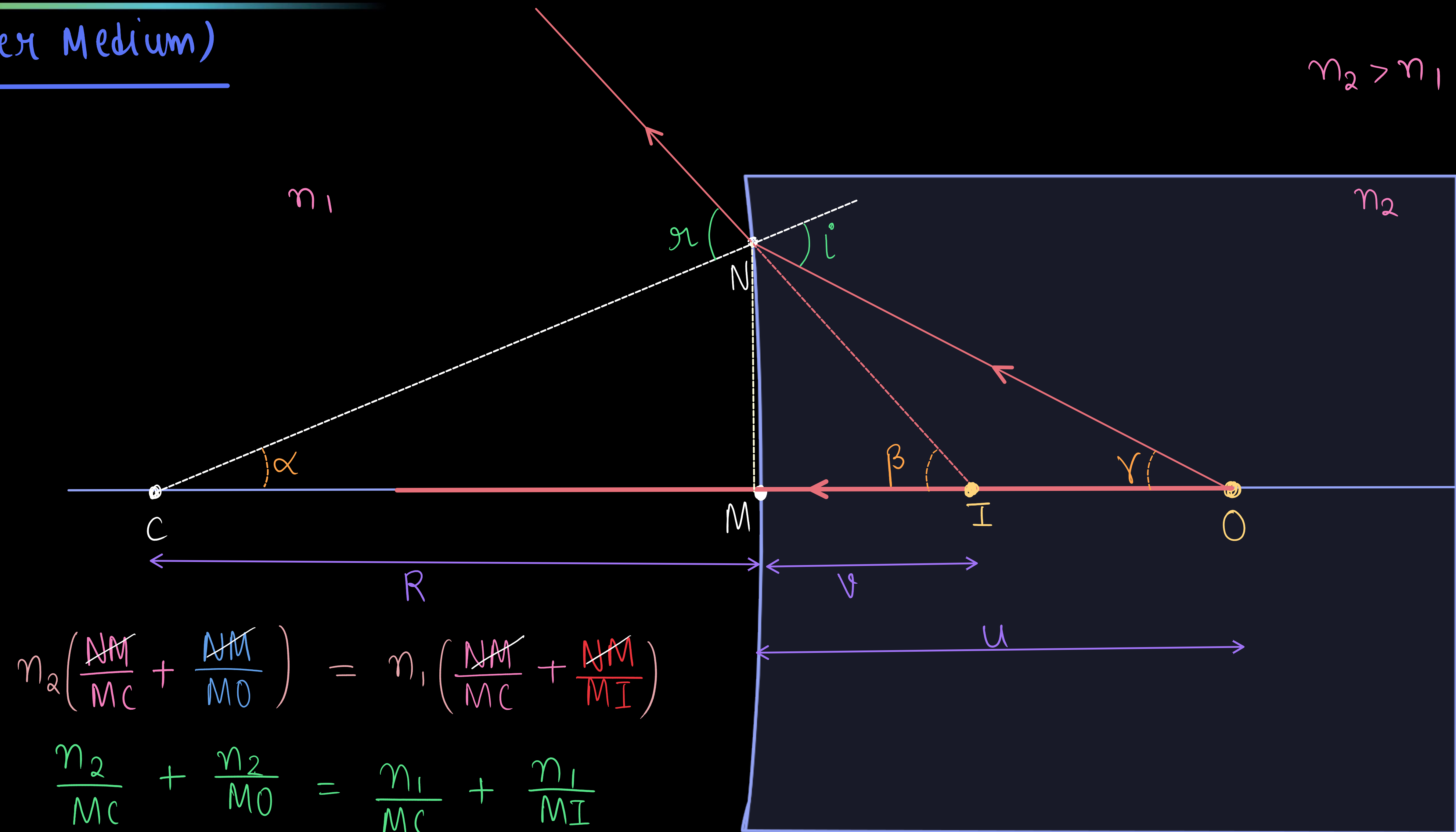
$$\tan \gamma \approx \gamma = \frac{NM}{MO} \quad \text{--- (5)}$$

using Snell's law -

$$n_2 \sin i = n_1 \sin r$$

$$\text{or } n_2 i = n_1 r \quad (\because \sin i \approx i, \sin r \approx r)$$

$$n_2(\alpha + \gamma) = n_1(\alpha + \beta)$$



$$n_2 \left(\frac{NM}{MC} + \frac{NM}{MO} \right) = n_1 \left(\frac{NM}{MC} + \frac{NM}{MI} \right)$$

$$\frac{n_2}{MC} + \frac{n_2}{MO} = \frac{n_1}{MC} + \frac{n_1}{MI}$$

$$\frac{n_2}{MO} - \frac{n_1}{MI} = \frac{n_1 - n_2}{MC}$$

using sign convention -

$$MO = -u, MI = -v, MC = R$$

$$\frac{n_2}{-u} - \frac{n_1}{-v} = \frac{n_1 - n_2}{R}$$

$$\boxed{\frac{n_1}{v} - \frac{n_2}{u} = \frac{n_1 - n_2}{R}}$$

Refraction at a spherical surface

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$

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Example 9.6 Light from a point source in air falls on a spherical glass surface ($n = 1.5$ and radius of curvature = 20 cm). The distance of the light source from the glass surface is 100 cm. At what position the image is formed?

Solution

We use the relation given by Eq. (9.16). Here

$u = -100$ cm, $v = ?$, $R = +20$ cm, $n_1 = 1$, and $n_2 = 1.5$.

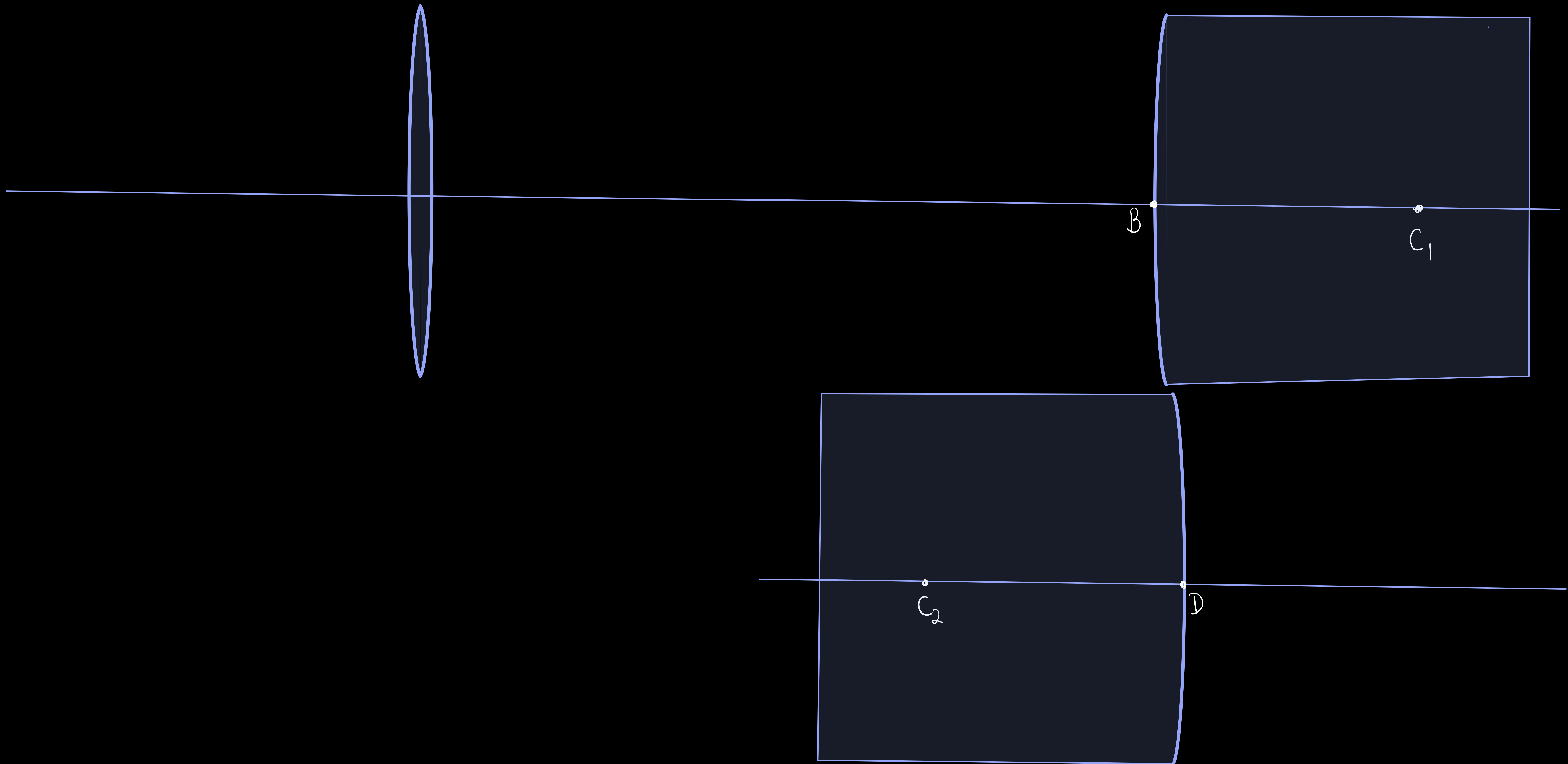
We then have

$$\frac{1.5}{v} + \frac{1}{100} = \frac{0.5}{20}$$

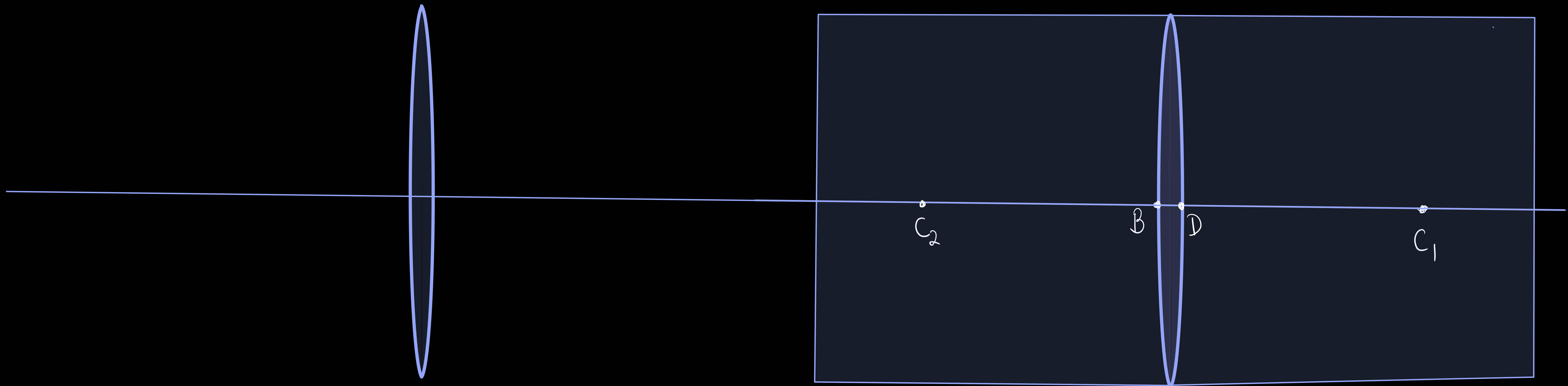
or $v = +100$ cm

The image is formed at a distance of 100 cm from the glass surface, in the direction of incident light.

Refraction by a lens (Lens Maker's Formula)

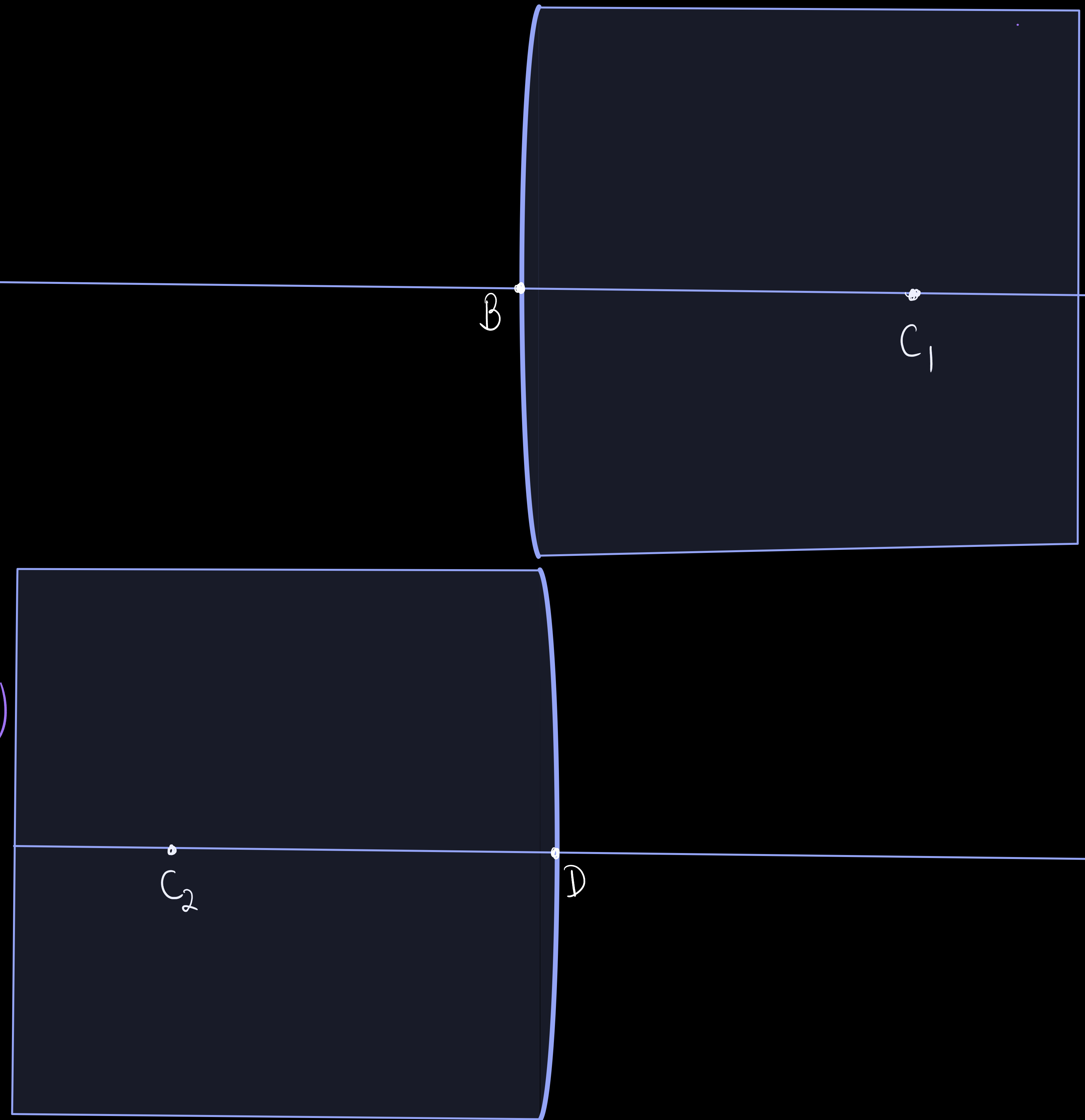
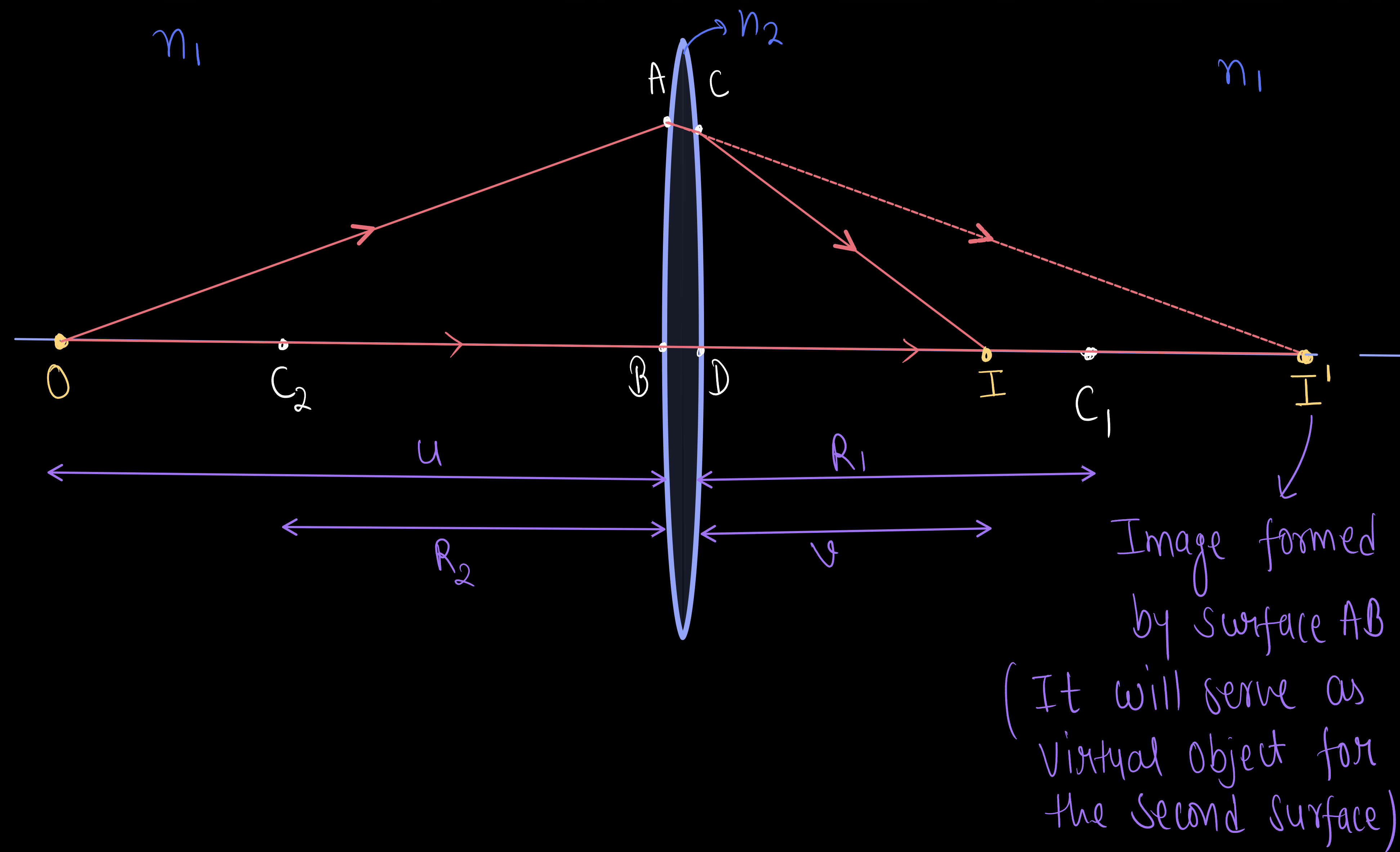


Refraction by a lens (Lens Maker's Formula)



- * A thin lens is a transparent optical medium bounded by two surfaces; at least one of which should be spherical.
- * Applying the formula for image formation by a single spherical surface successively at the two surfaces of a lens, we shall obtain the lens maker's formula and then the lens formula.

Refraction by a lens (Lens Maker's Formula)



Refraction by a lens (Lens Maker's Formula)

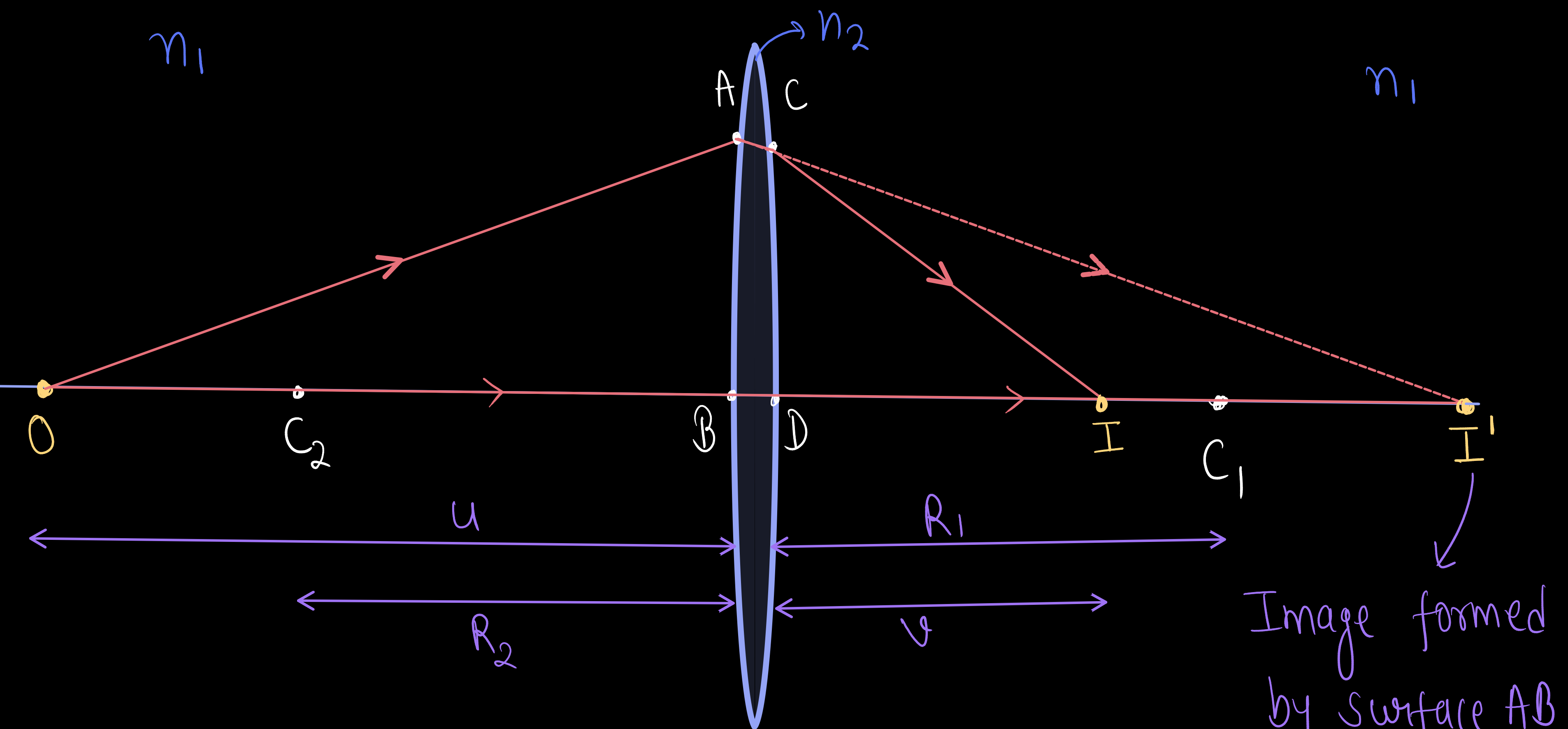
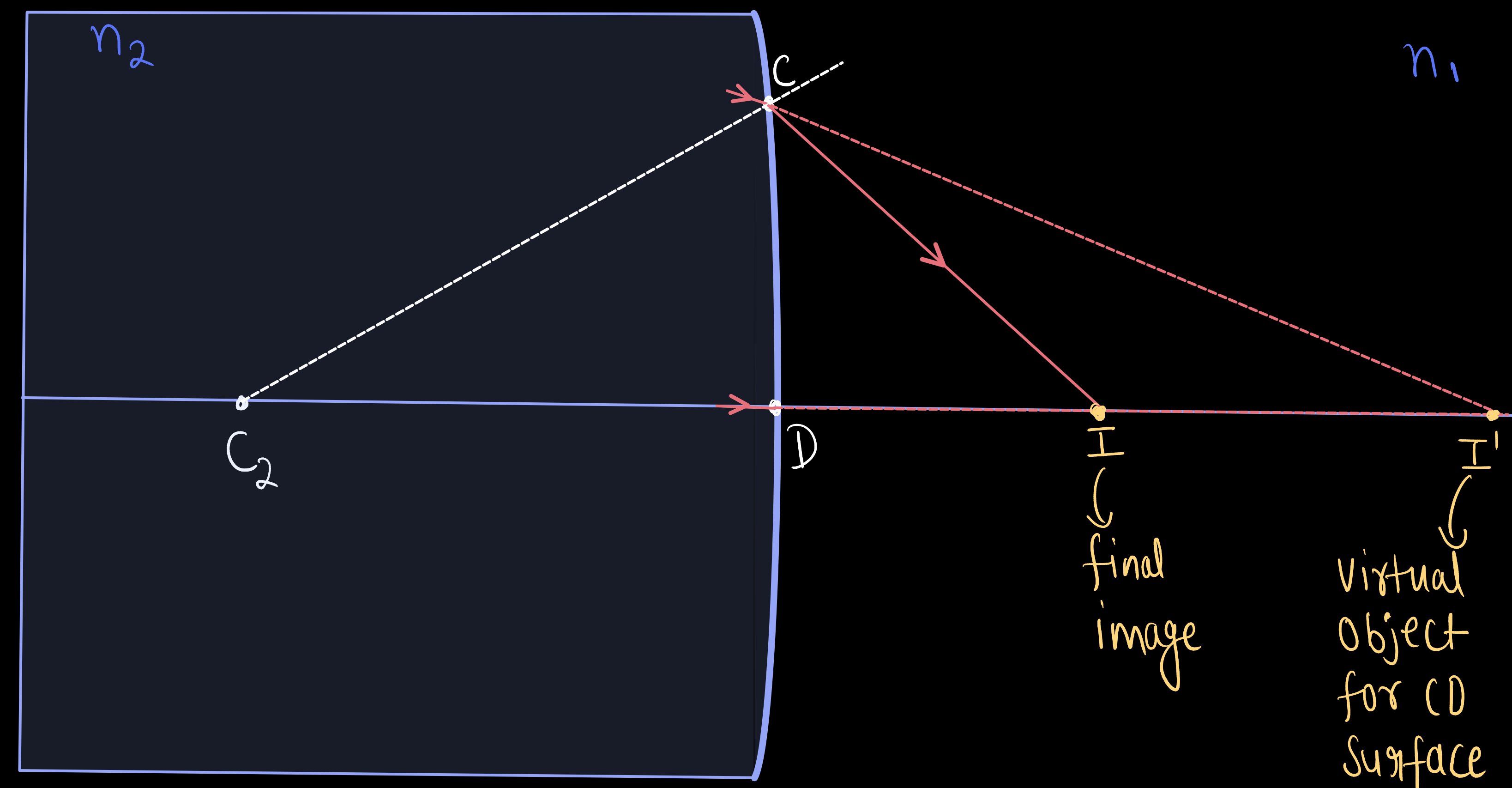
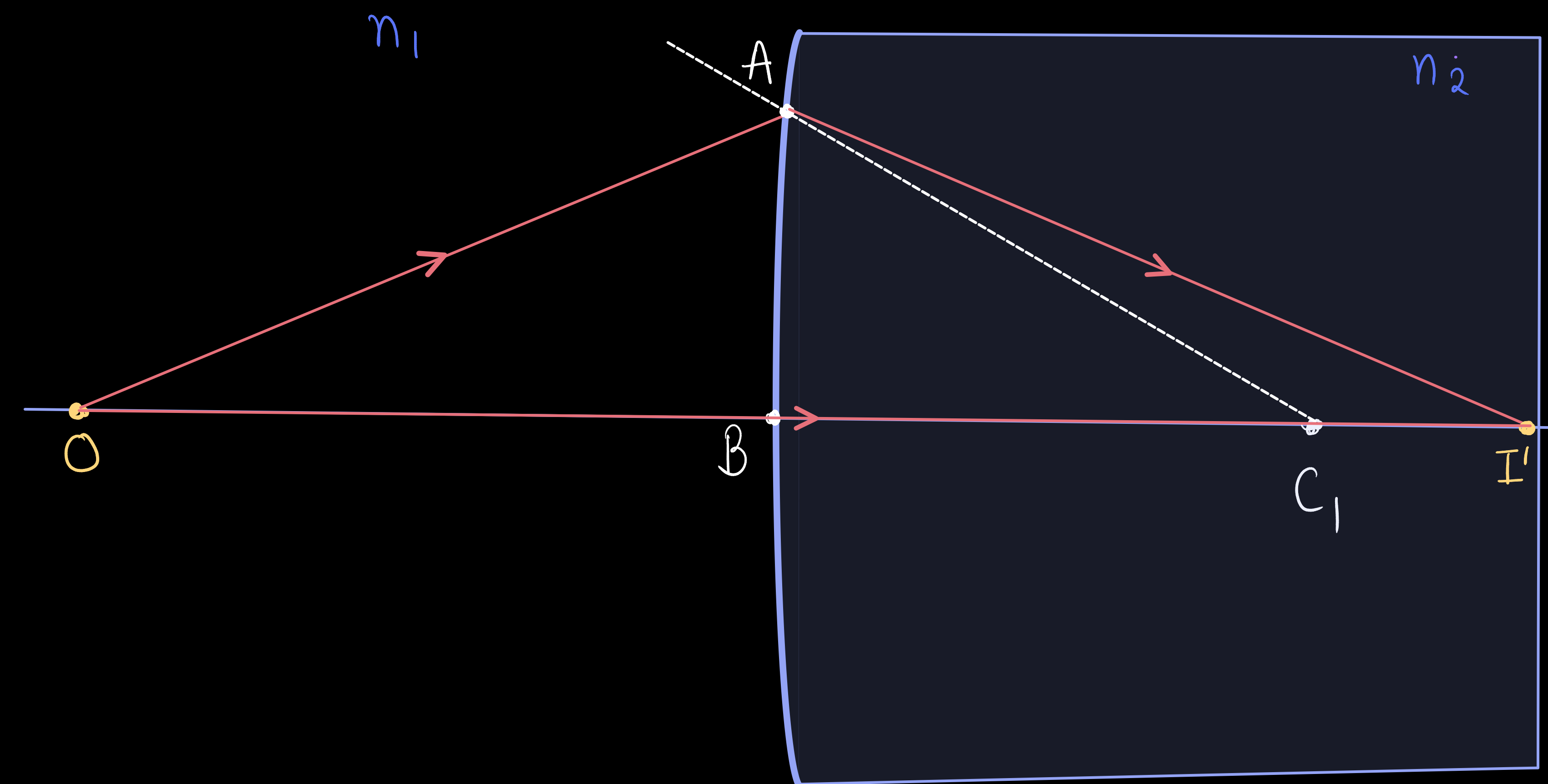


Image formed by surface AB
(It will serve as virtual object for the second surface)



* for first interface (AB) -

$$\frac{n_2}{BI'} - \frac{n_1}{(-BO)} = \frac{n_2 - n_1}{BC_1} \quad \text{--- (1)}$$

* for second interface (CD) -

$$\frac{n_1}{DI} - \frac{n_2}{DI'} = \frac{n_1 - n_2}{-DC_2} \quad \text{or}$$

$$\frac{n_1}{DI} - \frac{n_2}{BI'} = \frac{n_2 - n_1}{DC_2} \quad \text{--- (2)}$$

$\hookrightarrow (\because DI' \approx BI' \text{ for thin lens})$

Refraction by a lens (Lens Maker's Formula)

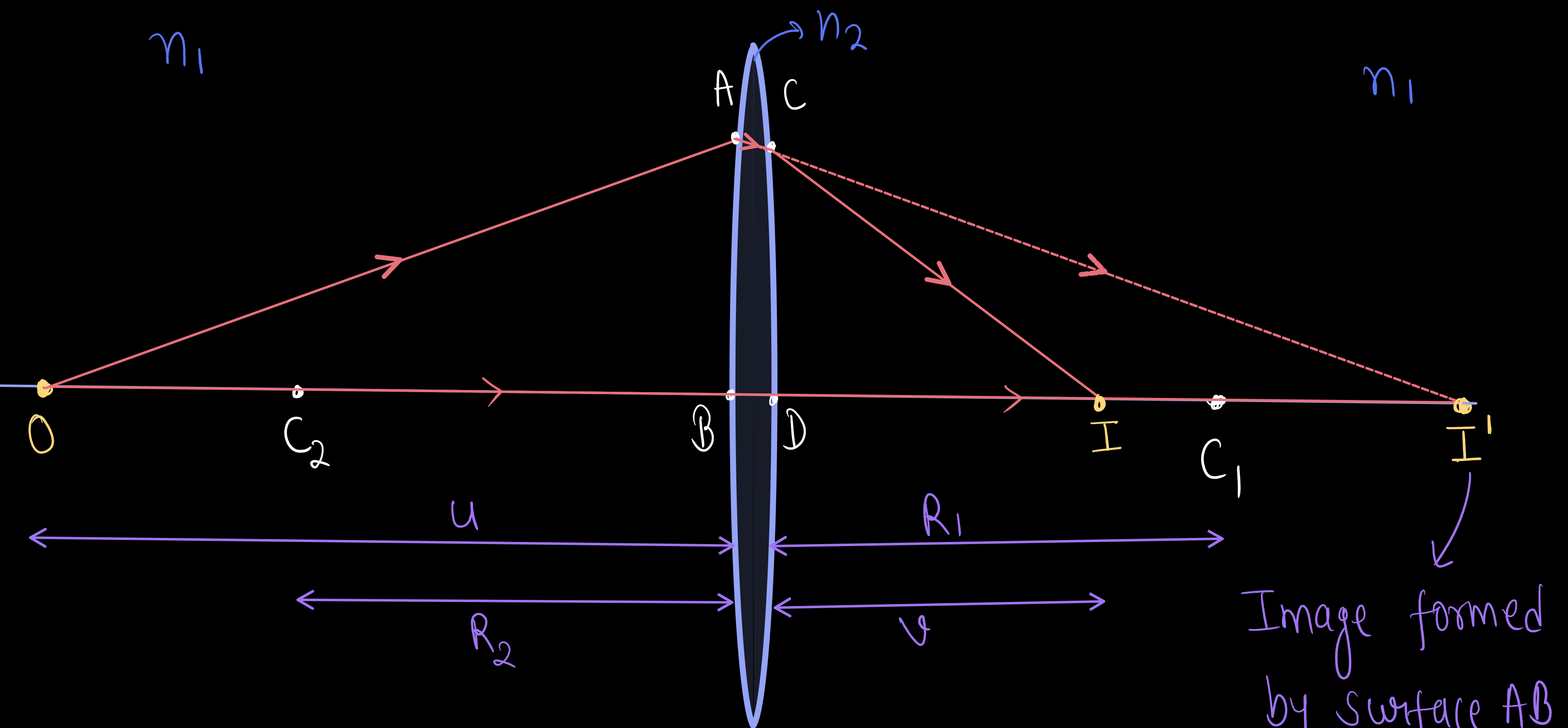


Image formed by surface AB
(It will serve as virtual object for the second surface)

* for first interface (AB) -

$$\frac{n_2}{BI'} - \frac{n_1}{(-BO)} = \frac{n_2 - n_1}{BC_1} \quad \text{--- (1)}$$

* for second interface (CD) -

$$\frac{n_1}{DI} - \frac{n_2}{DI'} = \frac{n_1 - n_2}{-DC_2} \quad \text{or} \quad \frac{n_1}{DI} - \frac{n_2}{BI'} = \frac{n_2 - n_1}{DC_2} \quad \text{--- (2)}$$

(∵ $DI' \approx BI'$ for thin lens)

Adding eq (1) & (2)

$$\frac{n_1}{DI} + \frac{n_1}{BO} = (n_2 - n_1) \left(\frac{1}{DC_1} + \frac{1}{BC_2} \right)$$

$$\frac{1}{DI} + \frac{1}{BO} = \left(\frac{n_2 - n_1}{n_1} \right) \left(\frac{1}{DC_1} + \frac{1}{BC_2} \right)$$

Now, using sign convention -

$$DI = v, \quad BO = -u, \quad DC_1 = R_1, \quad BC_2 = -R_2$$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad \text{--- (3)}$$

if $u = -\infty, v = f$

$$\frac{1}{f} - \frac{1}{\infty} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\boxed{\frac{1}{f} = \left(\frac{n_2}{n_1} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)} \quad \text{--- (4)} \quad \text{Lens maker's formula.}$$

from eq (3) & (4) $\Rightarrow$ $\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}} \quad \text{--- (5)} \quad \text{Lens formula.}$

Power of a lens

* Power of a lens is a measure of the convergence or divergence, which a lens introduces in the light falling on it.

* Clearly, a lens of shorter focal length bends the incident light more.

* The power P of a lens is defined as the tangent of the angle by which it converges or diverges a beam of light parallel to the principal axis falling at unit distance from the optical centre

$$P = \tan \delta = \frac{1}{f}$$

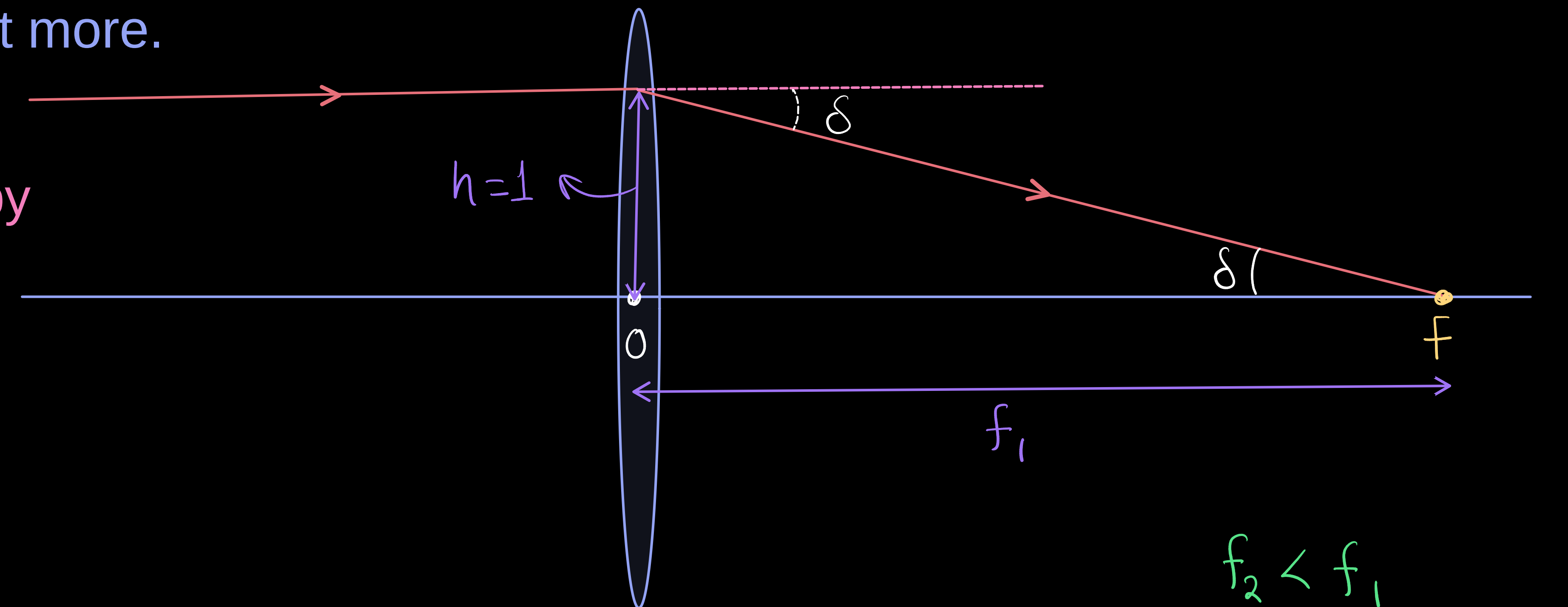
for small δ , $\tan \delta \approx \delta$

$$P = \delta = \frac{1}{f}$$

* The SI unit for power of a lens is diopetre (D) : $1D = 1m^{-1}$

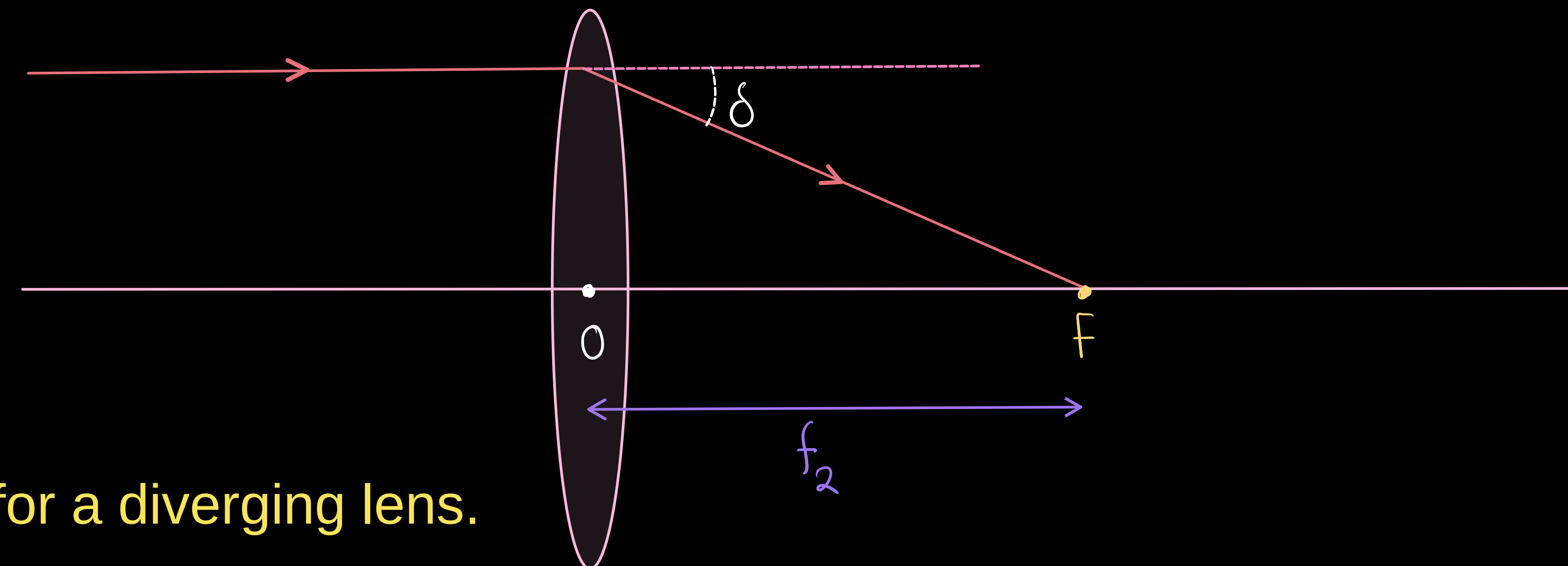
* The power of a lens of focal length of 1 metre is one diopetre.

* Power of a lens is positive for a converging lens and negative for a diverging lens.



$$f_2 < f_1$$

$$P_2 > P_1$$



Combination of thin lenses in contact

* Consider two lenses A and B of focal length f_1 and f_2 placed in contact with each other.

* The first lens A produces an image at I_1 .

* Since image I_1 is real, it serves as a virtual object for the second lens B, producing the final image at I.

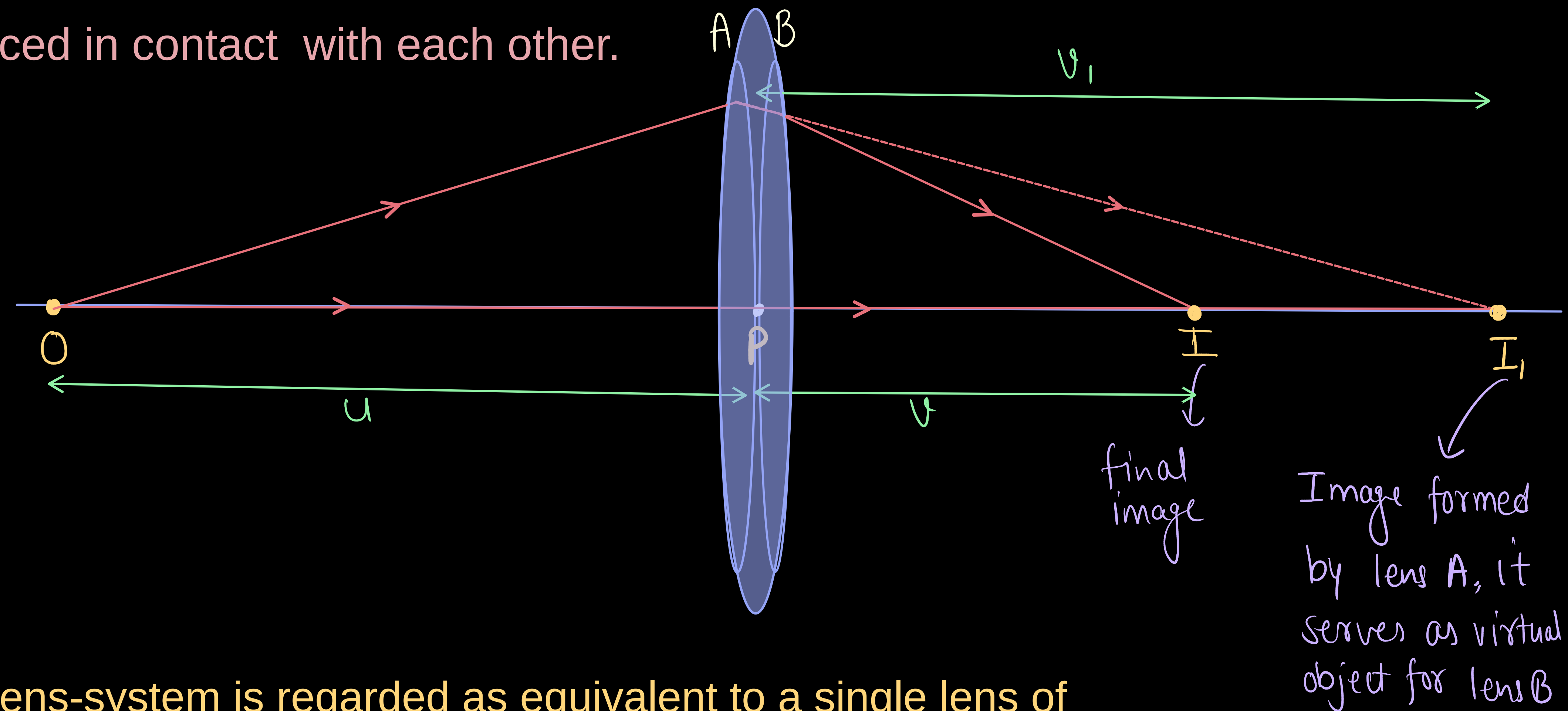


Image formed by first lens(A) -

$$\frac{1}{f_1} = \frac{1}{v_1} - \frac{1}{u} \quad \text{--- (1)}$$

Image formed by second lens(B) -

$$\frac{1}{f_2} = \frac{1}{v} - \frac{1}{v_1} \quad \text{--- (2)}$$

* Adding eq(1) and (2)

$$\frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{v} - \frac{1}{u} \quad \text{--- (3)}$$

* If the two lens-system is regarded as equivalent to a single lens of focal length f , we have

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} \quad \text{--- (4)}$$

from eq(3) and (4) we get

$$\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}} \quad \text{--- (5)}$$

* Power of Combined lens-

$$\boxed{P = P_1 + P_2} \quad \text{--- (6)}$$

* Total magnification (m) of the combination-

$$\boxed{m = m_1 \times m_2} \quad \text{--- (7)}$$

Example 9.9 Find the position of the image formed by the lens combination given in the Fig. 9.22.

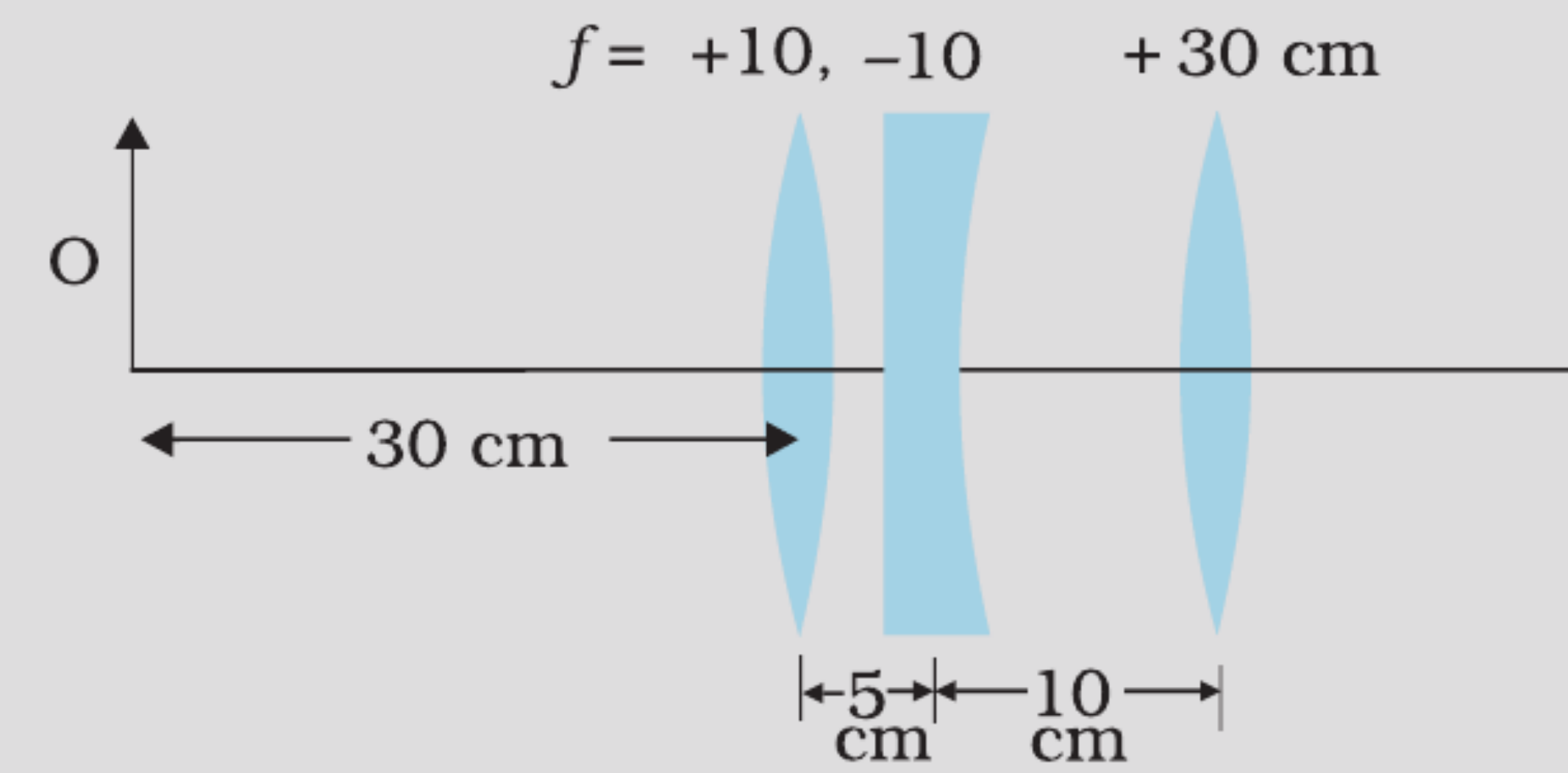


FIGURE 9.22

Solution Image formed by the first lens

$$\frac{1}{v_1} - \frac{1}{u_1} = \frac{1}{f_1}$$

$$\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{10}$$

$$\text{or } v_1 = 15 \text{ cm}$$

The image formed by the first lens serves as the object for the second. This is at a distance of $(15 - 5) \text{ cm} = 10 \text{ cm}$ to the right of the second lens. Though the image is real, it serves as a virtual object for the second lens, which means that the rays appear to come from it for the second lens.

$$\frac{1}{v_2} - \frac{1}{10} = \frac{1}{-10}$$

$$\text{or } v_2 = \infty$$

The virtual image is formed at an infinite distance to the left of the second lens. This acts as an object for the third lens.

$$\frac{1}{v_3} - \frac{1}{u_3} = \frac{1}{f_3}$$

$$\text{or } \frac{1}{v_3} = \frac{1}{\infty} + \frac{1}{30}$$

$$\text{or } v_3 = 30 \text{ cm}$$

The final image is formed 30 cm to the right of the third lens.